Chapter 1

Fundamentals of Bayesian Inference

Consider an observed dataset $\mathcal{D}$ and a statistical model parameterized by an unknown parameter vector $\boldsymbol{\theta} \in \Theta$, where Θ denotes the parameter space. In the Bayesian framework, uncertainty about $\boldsymbol{\theta}$ before observing the data is represented by a prior distribution $p(\boldsymbol{\theta})$.

Given a particular value of $\boldsymbol{\theta}$, the observed data are characterized by the conditional distribution

$$p(\mathcal{D} \mid \boldsymbol{\theta}),$$

which, when viewed as a function of $\boldsymbol{\theta}$ for fixed $\mathcal{D}$, defines the likelihood function.

The joint distribution of $\boldsymbol{\theta}$ and $\mathcal{D}$ can be factorized in two equivalent ways:

$$p(\boldsymbol{\theta}, \mathcal{D}) = p(\mathcal{D} \mid \boldsymbol{\theta})p(\boldsymbol{\theta}), \qquad p(\boldsymbol{\theta}, \mathcal{D}) = p(\boldsymbol{\theta} \mid \mathcal{D})p(\mathcal{D}).$$

Equating the two factorizations yields

$$p(\boldsymbol{\theta} \mid \mathcal{D})p(\mathcal{D}) = p(\mathcal{D} \mid \boldsymbol{\theta})p(\boldsymbol{\theta}),$$

and therefore Bayes' theorem gives

$$\boxed{p(\boldsymbol{\theta} \mid \mathcal{D}) = \frac{p(\mathcal{D} \mid \boldsymbol{\theta})p(\boldsymbol{\theta})}{p(\mathcal{D})}}. \tag{1.1}$$

The denominator is obtained by marginalizing the joint distribution over the parameter space:

$$\boxed{p(\mathcal{D}) = \int_{\Theta} p(\mathcal{D} \mid \boldsymbol{\theta})p(\boldsymbol{\theta})\mathrm{d}\boldsymbol{\theta}}. \tag{1.2}$$

The quantity $p(\mathcal{D})$ is referred to as the marginal likelihood, or model evidence. It normalizes the posterior distribution so that

$$\int_{\Theta} p(\boldsymbol{\theta} \mid \mathcal{D})\mathrm{d}\boldsymbol{\theta} = 1.$$

For a fixed observed dataset $\mathcal{D}$, the marginal likelihood $p(\mathcal{D})$ does not depend on $\boldsymbol{\theta}$. Hence, the posterior distribution can be written up to a proportionality constant as

$$\boxed{p(\boldsymbol{\theta} \mid \mathcal{D}) \propto p(\mathcal{D} \mid \boldsymbol{\theta}) p(\boldsymbol{\theta})}. \tag{1.3}$$

Thus, the basic principle of Bayesian inference can be summarized as

$$\boxed{\text{Posterior} \propto \text{Likelihood} \times \text{Prior}}.$$

Bayesian inference can also be used to quantify uncertainty about future observations. Let $\mathcal{D}^*$ denote a future observation or a set of future observations. By marginalizing over the uncertainty in $\boldsymbol{\theta}$, the posterior predictive distribution is

$$p(\mathcal{D}^* \mid \mathcal{D}) = \int_{\Theta} p(\mathcal{D}^* \mid \boldsymbol{\theta}, \mathcal{D}) p(\boldsymbol{\theta} \mid \mathcal{D}) \mathrm{d}\boldsymbol{\theta}.$$

If $\mathcal{D}^*$ is conditionally independent of $\mathcal{D}$ given $\boldsymbol{\theta}$, i.e., $\mathcal{D}^* \perp \mathcal{D} \mid \boldsymbol{\theta}$, then

$$p(\mathcal{D}^* \mid \boldsymbol{\theta}, \mathcal{D}) = p(\mathcal{D}^* \mid \boldsymbol{\theta}),$$

and the posterior predictive distribution reduces to

$$\boxed{p(\mathcal{D}^* \mid \mathcal{D}) = \int_{\Theta} p(\mathcal{D}^* \mid \boldsymbol{\theta}) p(\boldsymbol{\theta} \mid \mathcal{D}) \mathrm{d}\boldsymbol{\theta}}. \tag{1.4}$$

The posterior predictive distribution therefore propagates uncertainty in model the parameters into uncertainty about future observations by averaging the predictive distribution over the posterior distribution of $\boldsymbol{\theta}$.